

Activity 14: Subnetting practice

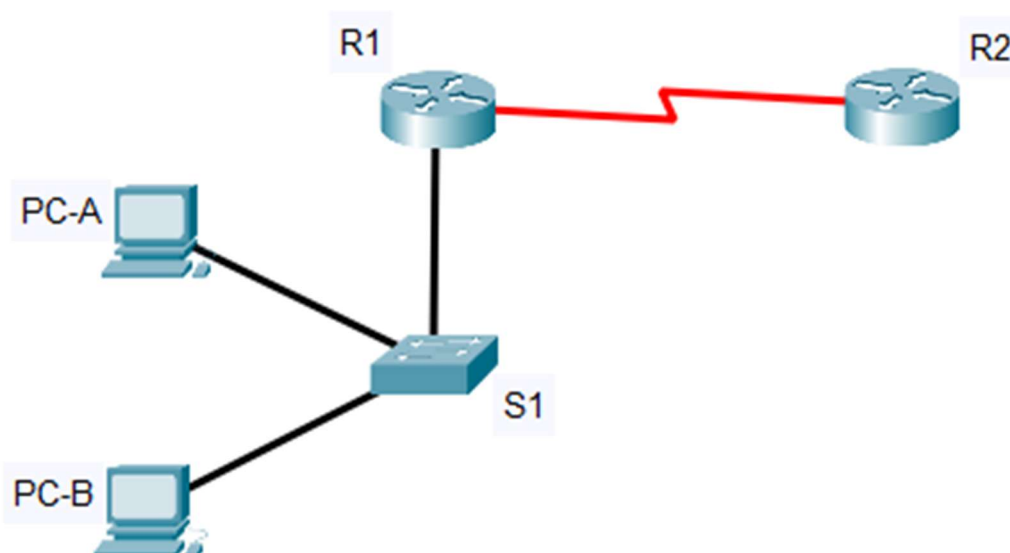
In this activity, you will examine a network diagram. Having the ability to calculate the necessary number of subnets when presented with a network architecture is crucial. Several scenario topologies, a base network address, and a mask will all be provided in this lab.

The network address will require being subnetted, and an IP addressing scheme will need to be proposed to support the number of subnets shown in the topology diagram. You are required to calculate the amount of bits to borrow, the number of hosts per subnet, and the possibility for expansion.

Instructions

Task 1: Identify the subnetting requirements for Topology 1

An overview of the entire network topology is illustrated in the diagram. You have been allocated the network address 192.168.40.0/24 to subnet, utilising the following topology for this task. After determining the required number of networks, create a suitable addressing scheme.



1. How many subnets are there in the network topology?

1. How many bits should you borrow to create the required number of subnets?

Use the subnetting formula:

$$2^n \geq \text{Number of required subnets}$$

Required subnets = **2**

$$2^1 = 2$$

Therefore: **Answer: Borrow 1 bit**

2. How many usable host addresses per subnet are in this addressing scheme?

With a **/25** mask:

- Host bits remaining = **7**
- Total addresses per subnet = $2^7 = 128$
- Usable host addresses = $128 - 2 = 126$

Answer: 126 usable host addresses per subnet

3. What is the new subnet mask in dotted decimal format?

255.255.255.128

4. How many subnets are available for future use?

You need **2 subnets** and borrowing **1 bit** creates exactly:

$$2^1 = 2 \text{ subnets}$$

Total subnets created = **2**

- Subnets currently used = **2**
- Future subnets available = **0**

Answer: 0 subnets available for future use

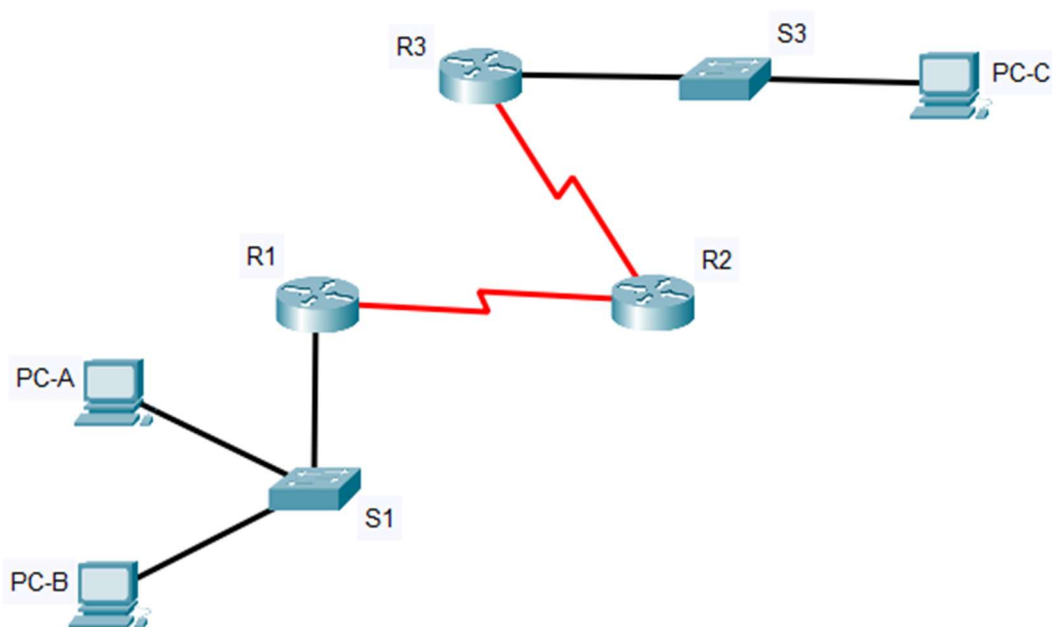
Fill in the following table with the subnet information:

Subnet number	Subnet address	First usable host address	Last usable host address	Broadcast address
1	192.168.40.0/25	192.168.40.1	192.168.40.126	192.168.40.127
2	192.168.40.128/25	192.168.40.129	192.168.40.254	192.168.40.255

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Task 2: Identify the subnetting requirements for Topology 2

The network topology from Task 1 has been expanded to accommodate the inclusion of a new components. Use the 192.168.40.0/24 network address to provide addresses to the network devices and then design a new addressing scheme to support the additional network requirement.



1. How many subnets are there?

4

2. How many bits should you borrow to create the required number of subnets?

Given network: **192.168.40.0/24**

Need at least **4 subnets**.

$$2^n \geq 4$$

$$2^2 = 4$$

Answer: Borrow 2 bits

3. How many usable host addresses per subnet are in this addressing scheme?

Original: **/24**

Borrow 2 bits: $/24 + 2 = /26$

Host bits remaining: $32 - 26 = 6$

Total addresses per subnet: $2^6 = 64$

Usable hosts: $64 - 2 = 62$

Answer: 62 usable host addresses per subnet

4. What is the new subnet mask in dotted decimal format?

A **/26** subnet mask is: **255.255.255.192**

Answer: 255.255.255.192

5. How many subnets are available for future use?

Borrowing 2 bits creates: $2^2 = 4$ subnets

Required subnets = **4**

Available for future use: $4 - 4 = 0$

Answer: 0 subnets available for future use

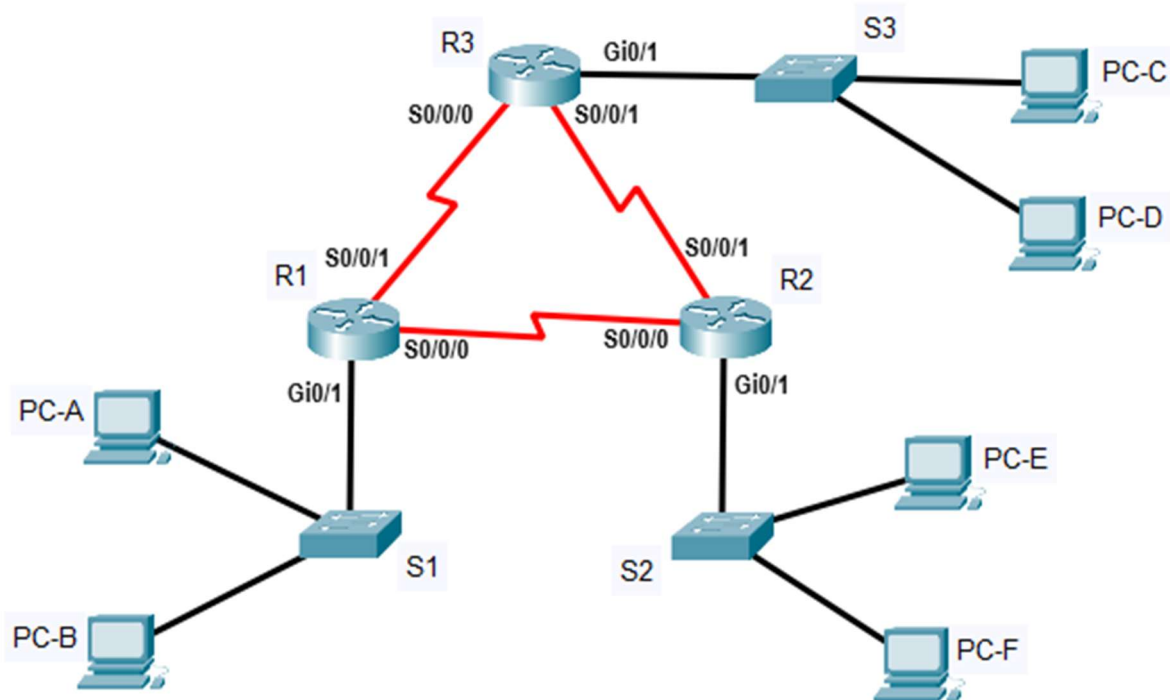
Fill in the following table with the subnet information:

Subnet number	Subnet address	First usable host address	Last usable host address	Broadcast address
1	192.168.40.0/26	192.168.40.1	192.168.40.62	192.168.40.63
2	192.168.40.64/26	192.168.40.65	192.168.40.126	192.168.40.127

3	192.168.40.128/26	192.168.40.129	192.168.40.190	192.168.40.191
4	192.168.40.192/26	192.168.40.193	192.168.40.254	192.168.40.255

Task 3: Identify the subnetting requirements for Topology 3: Challenge

The network topology from Task 2 has been expanded to accommodate the inclusion of new components with a new LAN added to R2, a redundant link between R1 and R3, and a new PC added to switch S3. Use the 192.168.40.0/24 network address to provide addresses to the network devices. Also provide an IP address scheme that will accommodate these additional devices. For this topology, assign a subnet to each network.



1. How many subnets are there?

6

2. How many bits should you borrow to create the required number of subnets?

Starting network: **192.168.40.0/24**

Need at least **6 subnets**.

Find the smallest number of borrowed bits n :

- $2^2 = 4$ (not enough)
- $2^3 = 8$ (enough)

Answer: Borrow 3 bits

3. How many usable host addresses per subnet are in this addressing scheme?

Borrowing 3 bits from a /24 network gives: **/27**

Host bits remaining: $32 - 27 = 5$

Total addresses per subnet: $2^5 = 32$

Usable host addresses: $32 - 2 = 30$

Answer: 30 usable host addresses per subnet

4. What is the new subnet mask in dotted decimal format?

A /27 subnet mask is: 255.255.255.224

5. How many subnets are available for future use?

Borrowing 3 bits creates: $2^3 = 8$ subnets

Subnets required: $8 - 6 = 2$

Answer: 2 subnets available for future use

Fill in the following table with the subnet information:

Subnet number	Subnet address	First usable host address	Last usable host address	Broadcast address
1	192.168.40.0/27	192.168.40.1	192.168.40.30	192.168.40.31
2	192.168.40.32/27	192.168.40.33	192.168.40.62	192.168.40.63
3	192.168.40.64/27	192.168.40.65	192.168.40.94	192.168.40.95

4	192.168.40.96/27	192.168.40.95	192.168.40.126	192.168.40.127
5	192.168.40.128/27	192.168.40.129	192.168.40.158	192.168.40.159
6	192.168.40.160/27	192.168.40.161	192.168.40.190	192.168.40.191
7	192.168.40.192/27	192.168.40.193	192.168.40.222	192.168.40.223
8	192.168.40.224/27	192.168.40.225	192.168.40.254	192.168.40.255

Fill in the following table with IP addresses and subnet masks for the router interfaces:

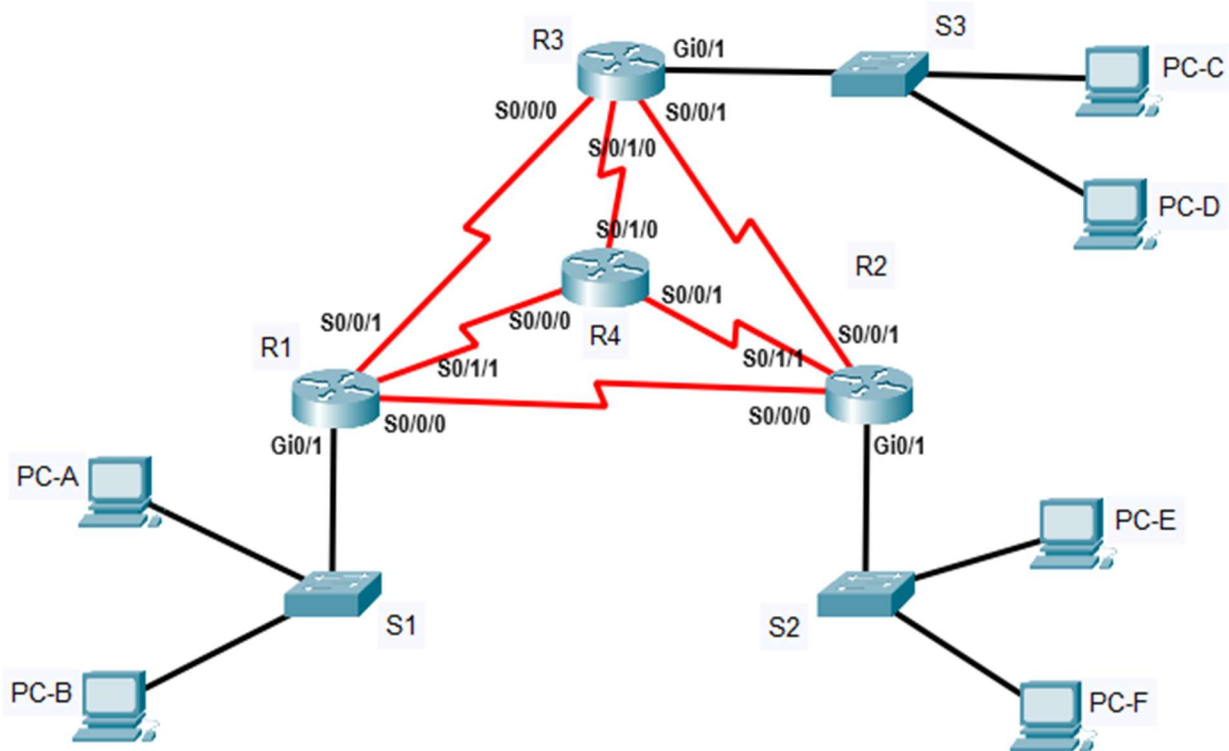
Device	Interface	IP address	Subnet mask
R1	Gi 0/1	192.168.40.1	255.255.255.224
	S 0/0/0	192.168.40.97	255.255.255.252
	S 0/0/1	192.168.40.129	255.255.255.252
R2	Gi 0/1	192.168.40.33	255.255.255.224
	S 0/0/0	192.168.40.98	255.255.255.252
	S 0/0/1	192.168.40.161	255.255.255.252
R3	Gi 0/1	192.168.40.65	255.255.255.224
	S 0/0/0	192.168.40.130	255.255.255.252
	S 0/0/1	192.168.40.162	255.255.255.252

Fill in the following table with the IP addresses and subnet masks for devices in the LAN as displayed in topology:

Device	IP address	Subnet mask	Default Gateway
PC-A	192.168.40.2	255.255.255.224	192.168.40.1
PC-B	192.168.40.3	255.255.255.224	192.168.40.1
PC-C	192.168.40.66	255.255.255.224	192.168.40.65
PC-D	192.168.40.67	255.255.255.224	192.168.40.65
PC-E	192.168.40.34	255.255.255.224	192.168.40.33
PC-F	192.168.40.35	255.255.255.224	192.168.40.33

Task 4: Identify the subnetting requirements for Topology 4: Challenge

The network was modified to accommodate changes in the organisation. The 192.168.40.0/24 network address is used to provide the addresses in the network.



1. How many subnets are there?

9

2. How many bits should you borrow to create the required number of subnets?

Starting network: **192.168.40.0/24**

Need at least **9 subnets**.

Find the smallest value of **n** such that: $2^n \geq 9$

$2^3 = 8$ ✗ not enough

$2^4 = 16$ ✓ enough

Answer: Borrow 4 bits

3. How many usable host addresses per subnet are in this addressing scheme?

Borrowing 4 bits from /24 gives: $/24 + 4 = /28$

Host bits remaining: $32 - 28 = 4$

Total addresses per subnet: $2^4 = 16$

Usable hosts: $16 - 2 = 14$

Answer: 14 usable host addresses per subnet

4. What is the new subnet mask in dotted decimal format?

A /28 subnet mask is: **255.255.255.240**

5. How many subnets are available for future use?

Borrowing 4 bits creates: $2^4 = 16$ subnets

Required subnets: $16 - 9 = 7$

Answer: 7 subnets available for future use

Fill in the following table with the subnet information:

Subnet number	Subnet address	First usable host address	Last usable host address	Broadcast address
1	192.168.40.0/28	192.168.40.1	192.168.40.14	192.168.40.15
2	192.168.40.16/28	192.168.40.17 -	192.168.40.30	192.168.40.31
3	192.168.40.32/28	192.168.40.33 -	192.168.40.46	192.168.40.47
4	192.168.40.48/28	192.168.40.49 -	192.168.40.62	192.168.40.63
5	192.168.40.64/28	192.168.40.65 -	192.168.40.78	192.168.40.79
6	192.168.40.80/28	192.168.40.81 -	192.168.40.94	192.168.40.95
7	192.168.40.96/28	192.168.40.97 -	192.168.40.110	192.168.40.111
8	192.168.40.112/28	192.168.40.113 -	192.168.40.126	192.168.40.127
9	192.168.40.128/28	192.168.40.129 -	192.168.40.142	192.168.40.143

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